

Stochastic Modelling of Catastrophic Storm Losses: Frequency-Severity Analysis for Insurance Pricing

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Abstract

Catastrophic events are happening more often than before. It creates major challenges for the insurance industry. One of the major issues is how to set premium prices correctly. Second is how to properly assess catastrophic risks. Using the data from the EM-DAT database, the thesis explains different stochastic methods to study the frequency and severity of storm-induced catastrophic events. Initially, Poisson distribution was opted for assessing the frequency of storm events; however, the violation of equidispersion was observed by the notably overdispersed data. As a result of variance exceeding the mean substantially, a Negative Binomial Distribution was better suited for estimating the frequencies of storm events.

Extreme Value Theory was employed for evaluating severity, to model the tail of the fitted storm losses data using the Generalized Pareto Distribution (GPD). The expected severity was determined to be infinite because of the heavy-tailed distribution of the data. This is infeasible in the context of traditional premium pricing.

With a given interest on insurance pricing and forecasting, the thesis is centered within the area of actuarial mathematics. Stressing on the role of the Extreme Value Theory (EVT) model in understanding catastrophic events. Also expanding on a method for estimating Pure Premiums in insurance pricing, namely the frequency-severity modelling.

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1 Introduction

1.1 Background and Context

Natural catastrophes are becoming more frequent due to climate change and economic development. Rapid urbanization and expansion of megacities have increased the frequency and severity of natural catastrophic events. Since the 1980s, the frequency of natural disasters has increased three folds and is expected to continue [7]. Continuous population and asset growth in risk-prone areas indicate that future risk will rise swiftly. The insurance sector is therefore under a lot of pressure to reduce these risks and hasten the recovery process after a disaster [3].

Natural catastrophes and increased future risk damage our society significantly. Damages can be direct, like environmental damages or indirect, such as financial losses. Insurance plays a major role in reducing financial burden by compensating households and businesses for these losses. However, an insubstantial proportion of these losses are covered by insurance globally. Furthermore, it is quite challenging to create an efficient insurance system for catastrophic risks. It is difficult to model and forecast catastrophic events using traditional actuarial techniques due to their rare and unpredictable nature. It is essential to have a strong and precise pricing model [7].

1.2 Key terms and Definitions

1. **Catastrophe Risk:** The possibility that extreme large and rare events, such as catastrophes, can result in substantial losses.
2. **Frequency:** The number of occurrences of storms during a specified time frame.
3. **Severity:** The amount of financial loss incurred due to a catastrophic event.
4. **Insurance Pricing:** The process of using risk assessment models to quantify the premium that policyholders must pay.

1.3 Current State of Knowledge

Climate and actuarial experts are actively investigating a fairly new research field of modelling climate risk insurance. The objective is to prepare the insurance industry competently for the consequences of catastrophe losses. Thereby, one has to understand that climate change can influence future risks. A fundamental component of the climate risk insurance model is accurately assessing present and future losses using probability, actuarial or catastrophe modelling.

1.4 Aim and Contribution

The aim of this thesis is to use cutting-edge stochastic models to evaluate storm-related disasters. This contributes to the expanding field of climate-induced risk modelling in insurance, making it relevant. To remain sustainable and efficient, insurers need to modify their pricing policies as per changes in catastrophic losses. This stresses on the need for data-driven pricing strategies via examining frequency and severity of storm events. These strategies can assist in the development of climate-resilient financial systems and disaster contingency plans [2].

2 Methodology

2.1 Overall Structure and Approach

This thesis was a representation of quantitative research that depended on existing historical data on storm events. The central aim was to estimate frequency and severity which can be further used for pricing insurance premiums. Regardless of the advancements in catastrophe modelling, significant gaps still exist. These gaps refer to the little use of statistical methods for measuring suitability and the lack of techniques to manage overdispersed data.

The research question used to guide this thesis and address such gaps is:

- Which statistical models are best suited to appropriately depict the frequency and magnitude of losses resulting from storm-related natural disasters?

2.1.1 Type of Data and Data Source

One of the most extensive and well-known disaster databases in the world, the EM-DAT International Disaster Database [5], provided quantitative secondary data for the study. With variables including disaster type, data, location, number of impacted individuals and projected financial losses, EM-DAT provides methodically gathered data on the incidence and effects of more than 22,000 major disasters globally.

The chosen dataset included various catastrophe events such as earthquakes, storms, wild-fires and floods along with their associated financial losses. When combined, the statistical behaviours, temporal patterns and reporting irregularities of many natural catastrophe events might make frequency-severity modelling more challenging. Thus, a decision was made to adopt a more homogeneous data by including only storm peril. Storms continuously had the greatest number of documented incidents with financial losses over the time period. This facilitated consistency, increased accuracy and credibility in model estimation. Furthermore, storms are one of the most common and economically significant hazards in the world, which makes them extremely important for evaluating insurance risk [4].

Existing historical data was chosen over experimental data due to the need for real-world, contextually relevant data. This is also recommended by prior catastrophic risk research [2]. Pricing models for catastrophe insurance are usually based on past loss data, which offers a reliable foundation for projecting future risk exposures [6]. The analysis was guaranteed to be both statistically powerful and globally relevant because of EM-DAT's extensive historical coverage and worldwide reach.

2.1.2 Quantitative Approach Justification

For a number of reasons, a quantitative approach was chosen. The primary study topics, which are best answered by statistical and probability models, first involved examining numerical data on the frequency of disasters and the magnitude of financial losses. Second, quantitative evaluations of loss distributions are crucial to the pricing of catastrophic premiums [3].

2.2 Data Collection and Preparation

2.2.1 Data Selection Process

Based on the requirements of the intended analysis, five criteria were developed when choosing data for this research:

1. **The year of occurrence:** allows for annual frequency counts and the identification of trends in the timing of disaster events.
2. **Catastrophe type:** restricted to natural catastrophes to eliminate man-made occurrences like industrial accidents, which have distinct risk dynamics and would skew the results of actuarial modelling.
3. **Catastrophe peril:** further defined to refer only to storm-related events, including cyclones, typhoons and hurricanes for the previously stated reasons.
4. **Insured and/or economic losses:** concentrating on the monetary losses incurred by each incident, since severity modelling requires financial loss data.
5. **Geographical and peril-specific consistency:** for comparability reasons, making sure that monetary values were expressed in US dollars.

Data from 24 years, from 2000 to 2023, were used in this study. The availability and trustworthy, regularly recorded data within EM-DAT for the designated peril and loss variables over these years was the main factor influencing this choice. Although certain disaster types have more comprehensive historical records in terms of number of occurrences, insured losses were only consistently accessible since the early 2000s.

To acquire a fair range of event variability and guarantee statistical reliability, it is generally advised from a methodological standpoint to employ at least 10 years of historic data for catastrophe modelling. Thus, using a 24 year timeframe not only meets the baseline requirement but also improves credibility of the calculated estimates. For modern catastrophe modelling, this longer timeframe makes it easier to identify new trends and long-term changes in catastrophe exposure and loss potential.

2.2.2 Data Measurement and Variables

This research measured two important variables, namely:

1. **Frequency:** The total number of storm-related disasters that occurred in 24 years.
2. **Severity:** The stated monetary losses (in US dollars) incurred during each storm event.

Data collection initially took into account both insured losses (the percentage of total losses covered by insurance) and economic losses (total losses resulting from a disaster, whether insured or not). However, only insured losses were included for severity analysis. This decision was necessary since insured losses directly relate to financial risk tackled by insurance firms. Uninsured losses, including infrastructure might be included in economic losses. In the context of insurance modelling, these losses provide less information. Therefore, insured losses were considered more focused as part of the dataset.

3 Data Analysis

The primary data analysis conducted to address the research goals mentioned in the previous chapter is presented in this chapter. The primary objective is to calculate frequency and severity of catastrophic losses. Since they have a direct impact on how insurance premiums are determined, these two elements serve as the cornerstone of actuarial pricing models; specifically, a fundamental actuarial principle called the frequency-severity analysis.

These estimates are combined to calculate the Pure Premium (sometimes called the Loss cost), which is the anticipated yearly cost of losses per unit of exposure on the assumption that exposure remains constant. It is computed as follows:

$$\text{Pure Premium} = \text{Expected Frequency} \times \text{Expected Severity} \quad (1)$$

When determining final premium rates, insurers base their calculations on the Pure Premium, which is the basic cost associated with covering the potential losses for policyholders. Other premium components, such as expenditures, profit loading and risk loading, are then added [1].

Python was the main tool used for data manipulation, statistical modelling and visualization in the analysis because of the amount and complexity of catastrophe loss datasets. There are a number of reasons for this. First off, Python has the capacity to effectively manage and analyse enormous and complex data structures without performance constraints, in contrast to traditional programs like Microsoft Excel that may find it difficult to handle and process large datasets. This was relevant to our work owing to the large catastrophe dataset containing many variables.

A vast array of specialised scientific and statistical libraries, including `NumPy`, `pandas`, `SciPy`, `Statsmodels`, `Scikit-learn`, and `Matplotlib`, are available in Python, as well. For statistical analysis, regression modelling, distribution fitting, diagnostic testing and visualization; all crucial elements of this study, these libraries provide reliable, user-friendly functionality. The implementation of sophisticated statistical models, such as the Extreme Value Theory and Negative Binomial Distribution, which are not easily supported in spreadsheet software settings, was made feasible by the availability of these tools.

Thirdly, Python improved the analysis' flexibility, transparency and reproducibility. Transparency in the generation of results was made possible by the ease with which all coding scripts, modelling procedures and visualization outputs could be recorded, examined and repeated. Python also made model comparisons, simulations and adjustments quicker and efficient, further improving the rigor of the research process.

3.1 Frequency Analysis

3.1.1 Exploratory Data Analysis

An exploratory data analysis (EDA) of the yearly count of storm events from 2000 to 2023 was the first stage in determining the frequency of storm events. Prior to using formal statistical models, this was essential for spotting early patterns, trends, and abnormalities in the data. To see how many storm events were documented annually over the course of a 24-year period, a time series plot was made.

Significant year-to-year fluctuation in storm frequencies was shown by the resulting plot Figure 1, with values ranging from as low as roughly 75 to over 130 incidents per year. During the reviewed period, there was no discernible long-term increase or decrease trend. However, there were evident short-term fluctuations over alternating periods.

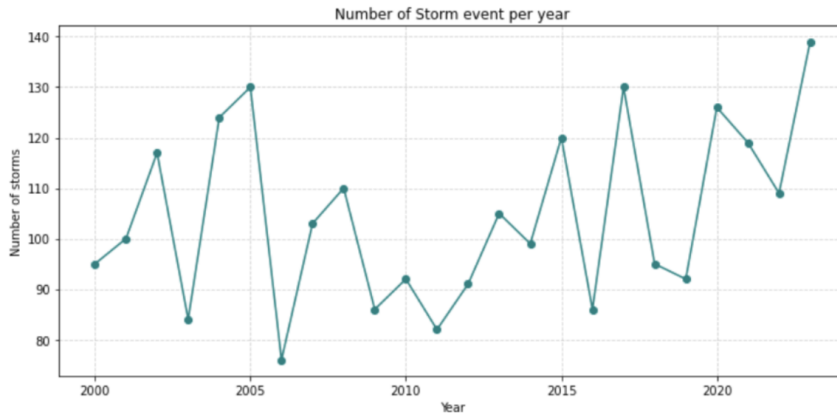


Figure 1: Annual number of storm events from 2000 to 2023

For example, clusters of above-average activity were observed in 2003-2005 and 2016-2017, but 2009-2011 were quieter years. Such differences suggest the possibility of greater climatic or environmental influences such as El Niño and La Niña patterns. However, these are outside the scope of this research.

Some years experienced unusually high storm event rates as shown in Figure 1. Notably, in 2005, 2017 and 2023, there were around 130 or more recorded storm occurrences. These were often followed by sharp declines, as shown by the significant decrease in events in 2006 and 2016. This pattern of occasionally outlier years interspersed with quieter periods is typical in catastrophe data and aligns with the cyclical nature of natural hazards.

In summary, the exploratory data analysis revealed important features of the storm frequency dataset. These included obvious annual fluctuations, strong years interspersed with slower ones and short-term clustering of active years.

3.1.2 Poisson Regression model

One of the most popular and advised models for examining count data is the Poisson distribution, especially when considering event occurrences across predetermined time periods [4]. It is a logical place to start when modelling the yearly frequency of storm-related disaster events because of its ease of use and proven statistical characteristics. However, its reliability and validity depends on the following key assumptions:

- **Event Independence:** Every catastrophic storm happens separately from the others, which means that the likelihood of one storm happening does not change when another does.
- **Constant Average Rate (λ):** Over time, the occurrences take place at a consistent mean rate. The average annual number of storm events is denoted by λ in this study.
- **Events Take Place One at a Time:** No two storms happen at the same time.
- **Equi-dispersion:** means that the variance of the event counts is equal to their mean.

Model Fitting and Implementation

After loading the dataset in Python, the total number of storm events annually for the 24-year period was determined. From this, the following formula was used to determine the average rate of occurrence each year (λ):

$$\lambda = \frac{\text{Total number of events}}{\text{Total number of years}} \quad (2)$$

Since there were 2510 storm events overall in the 24-year period, the estimated event rate was $\lambda = 104.58$. The Poisson Possibility Mass Function (PMF) was then used to describe the predicted frequency distribution of occurrences using Python's `scipy.stats` module. The Poisson PMF formula is as follows:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad (3)$$

where:

X : The random variable representing the number of occurrences (events) in a fixed interval.

k : The actual number of events occurring.

λ : The average rate (expected number) of occurrences.

To visualize the fit, a histogram of observed annual event counts was shown next to the Poisson-distributed data as shown in Figure 2.

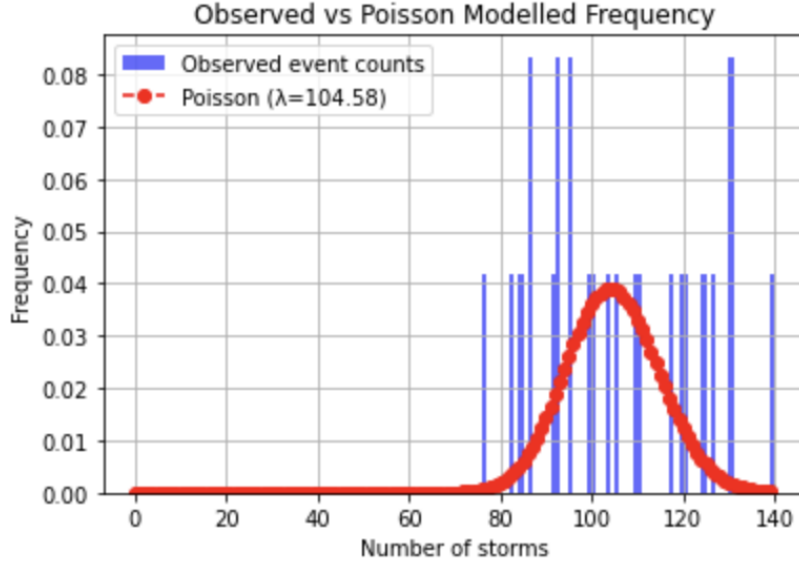


Figure 2: Comparison of Observed Annual Storm Event Counts and Fitted Poisson Distribution

First, the Poisson model fit the data's central tendency rather well, since it was accurately centered around the empirical mean of roughly 104.58 events annually. At the distribution's tails, however, the model showed a notably poor fit. In particular, it understated the number of exceptionally low event years (such as those with fewer than 90 events) and extremely high event years (such as those with more than 130 events). These differences were evident in the graph, where observed counts displayed sharp rises and falls in contrast to the Poisson model's flat bell-shaped curve.

Overdispersion is indicated by this mismatch in the distribution tails. The Poisson model implies independent and identically distributed storm events at a constant rate. However, this is not adequately account for the variability shown in real world scenarios such as catastrophe data. Thus, two statistical tests, the Chi-squared goodness-of-fit test and a dispersion test, were conducted in order to formally validate the existence of overdispersion and poor fit of the Poisson model.

Goodness-of-fit Test

A Chi-squared goodness-of-fit test was used to evaluate how well the poisson distribution suited the observed frequency data. Using Python's `scipy.stats.chisquare` function, the test contrasted the expected frequencies produced by the Poisson model (which is determined by multiplying the PMF by the number of years) with the observed frequency of events per year.

The Chi-squared statistic is computed as follows:

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} \quad (4)$$

where:

- O_i the observed frequencies.
- E_i : the expected frequencies.

A greater difference between the observed and expected frequencies is indicated by a higher Chi-squared statistic. The statistical significance of this disagreement is determined by the test's p-value, which is usually assessed against a common significance level of 0.05.

Because of the inherent rounding error in discrete probability distributions, one of the first problems was that the sum of expected frequencies did not equal the sum of observed frequencies. In order to remedy this, the predicted frequencies were rescaled and normalized using the following formula to make sure they added up to the total number of occurrences observed:

$$\text{Adjusted } E_i = E_i \times \frac{\text{Total } O_i}{\text{Total } E_i} \quad (5)$$

The Chi-squared test was run following the correction. The following were the outcomes:

- Chi-square statistic: 428.68
- p-value: 0.0000...

Notably, the p-value was incredibly low and far below the traditional threshold of 0.05, implying that it is statistically insignificant. The measured frequency of storm events varies greatly from what the Poisson distribution would predict. Thus, it rejects the null hypothesis by violating the Poisson assumption of equi-dispersion, ie: mean = variance. The Poisson regression model was proven to not be ideal for assessing frequency.

The Dispersion Test

A dispersion test was conducted to prove overdispersion of data. In the test, the predicted event rate (λ) was compared to the sample variance of event counts. The dispersion index was computed as follows:

$$\text{Dispersion Index} = \frac{\text{Variance of Event counts}}{\lambda} \quad (6)$$

The results are as follows:

- Mean (λ) = 104.58
- Variance = 310.51
- Dispersion index = 2.97

The Poisson assumption of equi-dispersion is clearly violated when overdispersion is confirmed by a dispersion index that is noticeably larger than 1. This is implied due to the fact that variance is significantly higher than the mean.

The occurrence of unusual, high impact events or clustering of catastrophic storms events within specific years could be one the potential reasons for this overdispersion. This pattern is frequently seen in catastrophe data, where some years have abnormally high number of storms because of underlying environmental or climatic reasons, while others are relatively calm. This clustering effect is inappropriate for modelling the frequency of such events since it deviates from the Poisson assumption of equi-dispersion and events taking place one at a time.

Conclusion and Model Progression

As stated before, the Poisson Regression model was concluded to be inappropriate for estimating the annual frequency of storm events in the dataset. This was confirmed by a low goodness-of-fit test and indications of overdispersion by the large dispersion index.

A Negative Binomial regression model was then used to give a more accurate and flexible depiction of the annual event frequency, in accordance with the best practices for managing overdispersed count data.

3.1.3 Negative Binomial Regression Model

The Negative Binomial (NB) distribution was thought to be a good substitute for the Poisson model due to the dataset's evidence of overdispersion, where the variance of storm event counts significantly outstripped the mean. This distribution is particularly suitable for modelling the frequency of catastrophe events and is frequently used in insurance modelling to manage overdispersed count data [4].

Negative Binomial distribution adds a second parameter to account for additional data variance. Thus, it provides a more adaptable structure than the Poisson distribution. The following formula represents the Negative Binomial distribution's probability mass function (PMF):

$$P(X = k) = \binom{k+r-1}{k} \cdot (p)^k \cdot (1-p)^r \quad (7)$$

where:

- $P(X = k)$: the probability of observing k events.
- r : the number of failures.
- p : the success probability.

The Negative Binomial distribution is a better fit for frequency modelling of catastrophe because it can decouple mean and variance.

The parameters r and p were estimated using the method of moments, which sets the theoretical mean and variance equal to the sample's values. Insurance modelling frequently uses this method, particularly when dealing with distributed data such as storm events. The formulas are shown below:

$$r = \frac{\mu^2}{\sigma^2 - \mu}, \quad p = \frac{\mu}{\sigma^2} \quad (8)$$

where:

- μ : mean of the distribution
- σ : variance of the distribution

Model Fitting and Implementation

Here, the same dataset was utilized as with the Poisson analysis. To guarantee that the predicted frequencies were calculated for the correct range and avoid calculation errors during the goodness-of-fit test, the range of event counts (from the lowest to maximum observed frequency) was determined.

The mean of 104.58 and variance of 310.51 were already known from the Poisson distribution. Using these values, the estimated parameters were calculated to be $p = 0.34$ and $r = 53.11$. Then, the predicted probabilities for each event count within this range

were calculated using Python's `scipy.stats.nbinom` function. Each probability was then multiplied by the total number of observations (years) to transform them into predicted frequencies.

Empty bins (event counts with zero observations) were dropped to account for possible divisions by zero and errors that may result from them. To guarantee a proper Chi-squared test comparison, the expected frequencies were also normalized to match the total observed frequencies, similar to the Poisson model. In order to visually confirm a better alignment than the Poisson model, a comparison plot showing observed against expected frequencies under the Negative Binomial model was then created, as shown below:

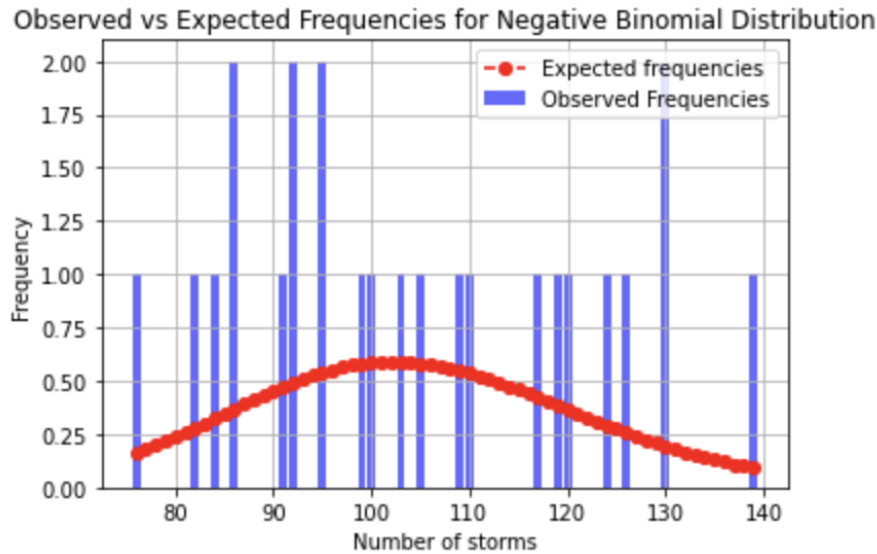


Figure 3: Observed vs Expected Event Frequencies for the Negative Binomial Distribution Fitted to Annual Storm Counts

In contrast to the Poisson model, there was significantly better alignment between the fitted Negative Binomial distribution and the observed yearly storm frequencies, particularly, in the lower and upper tails of the distribution. The data's greater variability and wider dispersion were considerably better represented in Figure 3. Both abnormally low and high storm counts were shown to have slightly higher probabilities by the Negative Binomial model, which more accurately represented the occurrence of exceptional years with either very few or extremely numerous storm events. In catastrophe modelling, this is crucial since infrequent but severe years can have disproportionately large financial effects on capital reserves and insurers.

The plot still showed a few specific years where the observed frequencies differed from the projected curve, even if the general fit was obviously better. These remaining disparities imply that, despite the Negative Binomial model's greater flexibility, storm frequency patterns may still be influenced by additional variables that a purely frequency-based model is unable to adequately account for. These variables can include regional clustering effects, multi-year climatic cycles or extreme climatic anomalies.

Goodness-of-fit Test

Similar to the Poisson model, a Chi-squared test was used to formally assess the model's adequacy. The outcomes were:

- Chi-square statistic: 74.6
- p-value: 0.15

The p-value of 0.15 is greater than significance level of 0.05. This suggests that there is no statistically significant difference between the observed frequencies and those predicted by the Negative Binomial model, indicating that we are unable to reject the null hypothesis at a standard significance level of 0.05. Thus, it implies that the annual frequency of storm events from the dataset can be satisfactorily fitted using the Negative Binomial regression model.

Conclusion

When compared to the Poisson model, the Negative Binomial model demonstrated a significantly better fit to the catastrophe frequency data, thereby accounting for the data's overdispersion. It is more popular for catastrophe modelling due to its ability to independently predict mean and variance.

As a result, our Negative Binomial regression model for predicting the expected frequencies of catastrophic events is now more accurate and statistically robust. In order to guarantee that insurers take into consideration the actual unpredictability and risk of storm occurrences when setting premiums, these improved predicted frequencies can be securely added to premium pricing models. Thus, it improves the financial stability of the insurance companies.

Quantifying the magnitude of monetary losses brought on by storm-related catastrophe occurrences is the next stage in determining the pure premium since it establishes the expected cost per claim in the case of a loss. The next section focuses on Extreme Value Theory for severity loss modelling since it has the ability to accurately depict the behaviour of rare but financially important catastrophe losses.

3.2 Severity Analysis

3.2.1 Exploratory Data Analysis

Extreme Value Theory, a subfield of statistics for modelling the behaviour of severe or rare events, was used to perform the severity analysis in this work. More specifically, the Peaks Over Threshold (POT) was used to model the dataset's exceedances above a chosen threshold using Generalized Pareto Distribution (GPD).

A number of important factors led to the choice to choose the POT method rather than the more conventional block maxima strategy. First, instead of depending only on the one maximum value inside each block or period, the POT approach records all exceedances beyond a predetermined high threshold. In catastrophe risk modelling, where each major loss event has significant financial ramifications, this makes it possible to retain more information from the extreme top tail of the distribution [2]. Additionally, compared to the overall number of storms documented, insured catastrophe loss records usually include very few extreme observations. In these situations, the POT approach is a viable and reliable option for estimating the magnitude of storm-related disaster losses since it is known to be statistically more effective and dependable for small sample sizes of extremes [4]

The choice of the Generalized Pareto Distribution (GPD) is theoretically supported by the Pickands-Balkema-de Haan theorem. The theorem asserts that for a wide class of underlying distributions, the distribution of values surpassing a sufficiently high threshold converges to a GPD, theoretically supports the use of the GPD for modelling these exceedances [4]. It explains why the GPD is appropriate for predicting the tail behaviour of insured loss data in this study and supports its usage in POT applications. The ability of the GPD to accurately depict the heavy-tailed character of catastrophe losses, where the biggest and most significant claims are clustered in the tail of the distribution, is another well-known feature. [4].

The key assumptions for Extreme Value Theory included:

- **Threshold Selection:** that is both high enough to identify excessive losses and retain enough data points for accurate estimation is crucial.
- **Assumption of Independence and Identically Distributed (i.i.d.):** The exceedances (insured loss amounts over the threshold) are assumed to be independent and identically distributed. This indicates that the statistical characteristics of losses do not change over time and that each extreme occurrence is independent of the others.
- **Assumption of Stationarity:** The distribution of extreme losses across time is assumed to be stationary under the EVT model. In other words, it is expected that the statistical features of high losses (such as the GPD's shape and scale parameters) remain constant during the study period.

Two essential parameters describe the distribution:

- **Shape parameter (ξ):** controls the spread and heaviness of the distribution.
- **Scale parameter (σ):** controls the tail behaviour of the distribution.

The Probability Density function (PDF) for the GPD is as follows:

$$f(x) = \frac{1}{\sigma} \left(1 + \xi \frac{x - \mu}{\sigma} \right)^{-\frac{1}{\xi} - 1}, \quad \text{for } \xi \neq 0 \quad (9)$$

The cumulative distribution function (CDF) for the GPD is given by:

$$F(x) = 1 - \left(1 + \xi \frac{x - \mu}{\sigma} \right)^{-\frac{1}{\xi}}, \quad \text{for } \xi \neq 0 \quad (10)$$

Expected severity, which is the expected size of losses that exceed the threshold, can be determined by the mean of this distribution.

3.2.2 Model Fitting and Implementation

First, Python was used to load the insured loss for storm events. Preprocessing required an appropriate threshold selection to identify which losses would be considered extreme. In EVT, setting a suitable threshold is essential because:

- An excessively low threshold would include many non-extreme losses, violating the GDP assumptions.
- Too high a threshold would leave too few exceedances for accurate statistical estimation.

Based on empirical testing and industry advice [4], the threshold was set at the 95th percentile of insured loss, called by the `np.percentile` function. The threshold was found to be at 5,000,000. A dataset of exceedances was created by extracting all insured loss values that exceeded the threshold found after it was established. After that, GPD was used to model these exceedances. Shape, location, and scale of the GPD were estimated using the `scipy.stats.genpareto.fit()` method. These estimates were found to be as follows:

- Shape parameter (ξ): 1.1018
- Scale parameter (σ): 8,162,979.71

The positive shape parameter validates the heavy-tailed distribution of insured losses, suggesting a higher likelihood of really large losses, which is in line with predictions for catastrophe data.

3.2.3 Threshold Diagnostic: Mean Residual Life (MRL) plot

A Mean Residual Life (MRL) Plot was created in order to confirm that the 95th percentile threshold was adequate. It is often called a Mean Excess Plot, commonly used to assess if data over a specified threshold can be accurately approximated by GPD. Graphically, the average excess values over different thresholds are plotted. Beyond a particular threshold, the plot should show an almost linear trend, indicating that the GPD is a good model for the exceedances [4]. Figure 4 shows the MRL plot for the insured losses.



Figure 4: Mean Residual Life (MRL) plot

A somewhat steady and linear trend was shown by the MRL plot, at an insured loss amount of US \$5,000,000. The plot showed obvious curvature and erratic variations below this threshold, indicating that losses in this region did not match the GPD's parameters. On the other hand, the mean excess values rose roughly linearly at the threshold of US \$5,000,000, which is in keeping with the behaviour predicted theoretically under a GPD. Thus, it was chosen as the ultimate threshold for the severity analysis based on this diagnostic result as well as the pragmatic considerations to balance the amount of exceedances available for accurate parameter estimation. This decision keeps the sample size large enough for reliable inference while guaranteeing that only very extreme storm loss occurrences are incorporated into the tail modelling.

To visually evaluate the accuracy of the GPD fit, a histogram of the empirical exceedances across the chosen threshold of US \$5,000,000 was created. Figure 5 shows how the fitted GPD closely matched the observed excess loss data's decaying trend. With a steadily declining likelihood of witnessing nigger excess losses, the fitted GPD showed heavy-tailed behaviour, which is typical of extreme loss data, with the majority of exceedances clustering around the lower excess values.

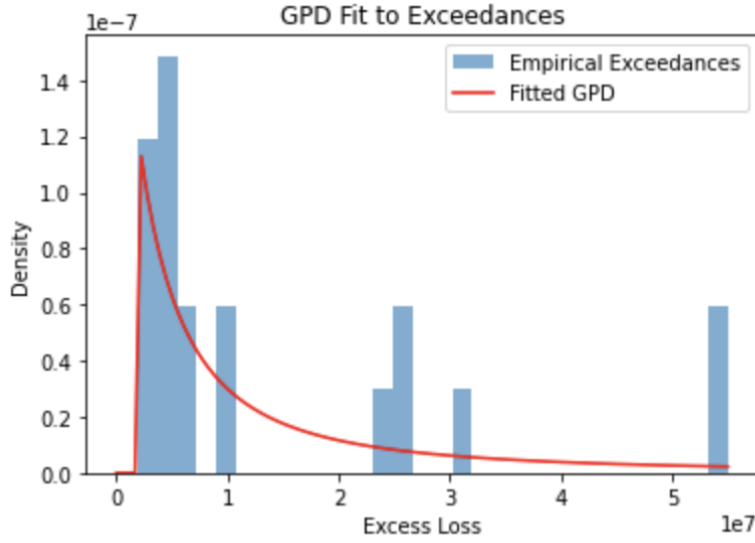


Figure 5: GPD fitted distribution against the empirical distribution

Although there were some differences between the empirical histogram and the fitted curve, particularly at specific loss levels where the empirical histogram bars momentarily exceeded the theoretical curve. This was to be expected because severe loss data is inherently sparse and unpredictable. Notwithstanding these little anomalies, the overall fit was acceptable and successfully depicted the loss distribution's fundamental tail behaviour.

3.2.4 Goodness-of-fit Test using Q-Q plot

The GPD fit was further assessed by creating a quantile-quantile (Q-Q) plot. The quantiles of the actual data are frequently compared to the theoretical quantiles from a given distribution, in this example the fitted GPD, using a graphical diagnostic tool called a Q-Q plot. The plotted points should ideally lie along a 45-degree reference line, signifying a tight match between the empirical and theoretical distributions, if the selected distribution matches the data well.

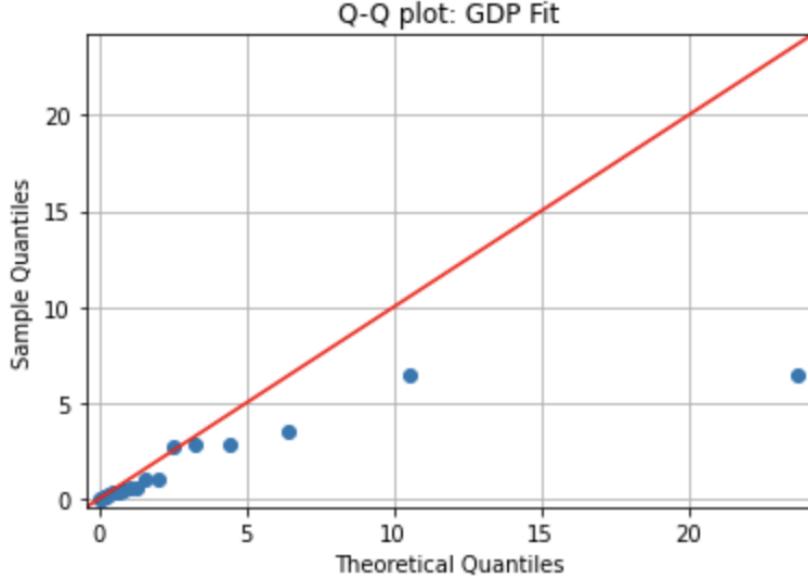


Figure 6: Q-Q plot for assessment of GPD fitted distribution

Figure 6 shows the Q-Q plot at the chosen threshold, revealing useful trends about the model's functionality. The data points closely matched the reference red line in the lower and moderate quantiles (left and center portions of the plot). This indicated that the GPD offered a good approximation for the majority of the excess loss values.

However, the points started to stray from the 45-degree reference line at the higher quantiles (towards the right end of the plot), especially for the biggest losses that were seen. The rarity, irregularity and intrinsic volatility of big loss events are the main causes of this difference. It is a common occurrence in extreme value modelling. Although significant, these deviations do not always compromise the overall adequacy of the model. This is because, like the majority of the extreme value models, GPD is generally better at modelling the moderate tail and body of the exceedance distribution than they are at accurately fitting the rarest, most extreme observations.

3.2.5 Expected Severity Estimation

Following the GPD process, the expected severity of exceedances was computed. The formula for expected severity in the GPD is given by:

$$E[X] = \begin{cases} u + \frac{1-\xi}{\sigma}, & \text{if } \xi < 1 \\ \infty, & \text{if } \xi \geq 1 \end{cases} \quad (11)$$

But since we typically work with excesses, $Y = X - u$, the expected severity beyond the threshold is [4]:

$$E[Y] = \begin{cases} \frac{\sigma}{1-\xi}, & \text{if } \xi < 1 \\ \infty, & \text{if } \xi \geq 1 \end{cases} \quad (12)$$

where ξ is the shape parameter, and σ is the scale parameter.

As stated before, the estimated shape parameter was found to be 1.1018 which is greater than 1. Thus, the expected severity becomes infinite when the shape parameter exceeds 1, as suggested by the formula above [4]. This suggests a heavy-tailed distribution, in which the expected value theoretically diverges and the likelihood of witnessing really big losses does not decrease quickly.

Given that natural catastrophes have the ability to produce rare but extremely severe events, such a result is not of the ordinary in catastrophe modelling. An infinite predicted severity implies that using a finite mean for risk measures or pricing is no longer feasible and may even be deceptive. To address the limitations of mean-based estimates, insurers frequently use quantile-based risk metrics like Conditional Tail Expectation (CTE) or Value-at-Risk (VaR), which concentrate on the possible magnitude of extreme losses at a specific confidence level. These risk metrics are often used tools in disaster risk management because they offer a more solid foundation for pricing and capital allocation choices under heavy-tailed loss distributions.

Conclusion

In summary, GPD was demonstrated to accurately depict the tail behaviour of loss distribution using exploratory analysis. This was proven by the goodness-of-fit assessment by using Q-Q plots. Nevertheless, a heavy-tailed distribution with an infinite expected severity was suggested by an estimated shape parameter greater than 1. Thus, it hindered the calculation for pure premiums for this dataset. This underscored the impracticality of mean-based loss estimations for catastrophe pricing.

4 Final Conclusion

This study focused on the frequency and severity of storm events and losses during a 24-year period to quantify pure premiums for insurance pricing. Given the increasing intensity of storms and the corresponding financial threats, it is essential for the insurance industry to develop robust pricing strategies that capture these rare, extreme events. By evaluating the storm events using different statistical distributions, this study contributes that goal.

One of the most popular starting points for modelling, namely Poisson distribution, was used to examine the frequency of storm events. Diagnostic testing, however, showed a substantial overdispersion in the data. Thus, it defied the equidispersion assumption of the Poisson model. With a variance significantly higher than the mean, the dispersion index was determined to be 2.9. In order to effectively account for the data's overdispersion, a Negative Binomial distribution was adopted. This led to a better fit of the observed frequency to the modelled frequencies.

Extreme Value Theory (EVT), more especially the Generalised Pareto Distribution (GPD) for losses over a predetermined threshold, was then used to analyse the severity of the insured storm losses. The results suggested an infinite expected severity for the losses as the shape parameter was greater than 1. Given the heavy-tailed nature of disaster loss distributions, this meant that the projected severity of storm losses was mathematically undefinable. As a result, one of the fundamental difficulties faced was that the pure premiums could not be computed from this information.

Since insured losses are covered for by the insurance companies, the research was limited to insured losses rather than total economic losses, which also take into account the uninsured losses. Although the study highlighted the effectiveness of statistical distributions in catastrophe modelling, it is important to recognize some of its limitations. The use of historical data may not adequately reflect the increasing and changing nature of climate-induced hazards since it is essentially based on the idea that past loss patterns are predictive of future ones.

Future research directions may include climate prediction models by combining the temporal variables or climatic anomalies. It is essential to investigate different severity distributions that can generate finite expectations for heavy-tailed data. Furthermore, under various climatic assumptions, the portfolio-risk might be evaluated using Monte-Carlo simulations and scenario-based modelling.

In conclusion, this study highlights the challenges and possibilities associated with catastrophe modelling for insurance pricing. Through the use of adaptable statistical techniques, the study emphasised on the need of actuarial models to evaluate climate data. The results underscore the adoption of sophisticated, adaptive models by the insurance industry to remain persistent against these increasingly severe catastrophes.

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5 Appendix

5.1 Python Code for Poisson process

```
1 # Cell 1
2 #Importing libraries to be used
3 import numpy as np
4 import pandas as pd
5 import matplotlib.pyplot as plt
6 from scipy.stats import poisson
7 from scipy import stats
8 from scipy.stats import nbinom, chisquare
9
10 # Cell 2
11 #Loading dataset
12 file_source = 'Catastrophe Perils Data.xlsx'
13 df = pd.read_excel(file_source, sheet_name='Storms')
14
15 #Preview of data
16 print(df.head())
17
18 #Number of storm events per year (time series line plot)
19 plt.figure(figsize=(10,5))
20 plt.plot(df['Year'], df['Event Frequency'], marker='o', color='teal')
21 plt.title('Number of Storm event per year')
22 plt.ylabel('Number of storms')
23 plt.xlabel('Year')
24 plt.grid(True, linestyle='--', alpha=0.6)
25 plt.tight_layout()
26 plt.show()
27
28 #Finding total number of events and years
29 total_years = df.shape[0]
30 total_events = df['Event Frequency'].sum()
31
32 #Calculating the average rate of occurrence per year
33 est_lambda = total_events / total_years
34 print(f"\nEstimated (average events per year): {est_lambda:.2f}")
35
36 #Plotting observed event frequency vs expected Poisson
    probabilities
37 max_event = df['Event Frequency'].max()
38 event_range = np.arange(0, max_event+1)
39
40 #Calculating Poisson PMF for range
41 poisson_pmf = poisson.pmf(event_range, est_lambda)
42
43 #Plotting histogram of observed counts
44 plt.hist(df['Event Frequency'], bins=range(0, max_event+2), density
    =True, alpha=0.6, color='blue', label='Observed event counts')
45
46 #Plotting Poisson distribution
47 plt.plot(event_range, poisson_pmf, 'r--', marker='o', label=f'
    Poisson (={est_lambda:.2f})')
```

```

48
49 plt.title('Observed vs Poisson Modelled Frequency')
50 plt.xlabel('Number of storms')
51 plt.ylabel('Frequency')
52 plt.legend()
53 plt.grid(True)
54 plt.show()
55
56 #Cell 3
57 #Goodness-of-fit (Chi-square)
58 obs_freq, bins = np.histogram(df['Event Frequency'], bins=range(0,
    max_event+2))
59 exp_freq1 = poisson.pmf(event_range, est_lambda) * total_years
60
61 #Adjusting the expected frequencies to match total observed data
62 exp_freq = exp_freq1 * (obs_freq.sum() / exp_freq1.sum())
63
64 #Performing the test
65 chi_square_stat, p_value = stats.chisquare(f_obs=obs_freq, f_exp=
    exp_freq)
66 print(f"Chi-square statistic: {chi_square_stat:.2f}")
67 print(f"p_value: {p_value:.12f}")
68
69 if p_value > 0.05:
70     print("Poisson distribution is a good for for the data")
71 else:
72     print("Poisson distribution is not ideal for the data")
73
74 #Cell 4
75 var = df['Event Frequency'].var()
76 dispersion = var / est_lambda
77 print(f"\nDispersion Analysis:")
78 print(f"Variance: {var:.2f}")
79 print(f"Mean: {est_lambda:.2f}")
80 print(f"Dispersion Index: {dispersion:.2f}")
81
82 if dispersion == 1:
83     print("Data follows Poisson distribution")
84 elif dispersion > 1:
85     print("Data is overdispersed")
86 else:
87     print("Data is underdispersed")

```

5.2 Python Code for Negative Binomial Distribution

```
1 #Cell 1
2 #Data is already loaded above
3 #Extracting values for event frequency
4 freq = df['Event Frequency'].values
5
6 #Calculating mean and variance
7 mean = np.mean(freq)
8 var = np.var(freq, ddof=1)
9
10 #Estimating parameters for Negative Binomial distribution
11 p = mean / var
12 r = mean**2 / (var-mean)
13
14 print(f"Mean: {mean}")
15 print(f"Variance: {var}")
16 print(f"Estimated p (success probability): {p}")
17 print(f"Estimated r (number of failures): {r}")
18
19 #Creating bins from min to max count (freq)
20 min_freq = min(freq)
21 max_freq = max(freq)
22 bins = np.arange(min_freq, max_freq+1)
23
24 #Finding observed and expected frequencies
25 observed_f = np.array([np.sum(freq==i) for i in bins])
26 expected_p = nbinom.pmf(bins, r, p)
27 expected_f = expected_p * len(freq)
28
29 #To avoid division by zero, eliminating bins where both observed
    and expected frequencies are zero
30 mask = (observed_f + expected_f) > 0
31 observed_f = observed_f[mask]
32 expected_f = expected_f[mask]
33
34 #Scaling expected frequencies to match the total number of
    observed frequencies
35 expected_f = expected_f * (observed_f.sum() / expected_f.sum())
36
37 #Plotting observed vs expected frequencies
38 plt.bar(bins, observed_f, color='blue', alpha=0.6, label='Observed
    Frequencies')
39 plt.plot(bins, expected_f, 'ro--', label='Expected frequencies')
40 plt.xlabel('Number of storms')
41 plt.ylabel('Frequency')
42 plt.title('Observed vs Expected Frequencies for Negative Binomial
    Distribution')
43 plt.legend()
44 plt.grid(True)
45 plt.show()
46
47 #Cell 2
48 chi2_stat, p_value = chisquare(f_obs=observed_f, f_exp=expected_f)
49 print(f"Chi-square statistic: {chi2_stat:.2f}")
50 print(f"p_value: {p_value:.4f}")
```

```
51 |
52 | if p_value < 0.05:
53 |     print("Negative Binomial distribution may not be a good fit for
54 |         the data")
55 | else:
56 |     print("Negative Binomial distribution is a good fit for the
57 |         data")
```

[language=Python]

5.3 Python Code for Extreme Value Theory

```
1 # Cell 1
2 import pandas as pd
3 import numpy as np
4 import matplotlib.pyplot as plt
5 from scipy.stats import genpareto
6 from statsmodels.graphics.gofplots import qqplot
7
8 #Cell 2
9 #Loading the dataset
10 df = pd.read_excel('Catastrophe Perils Data.xlsx', sheet_name='
    Sheet1')
11
12 #Renamed column to an easier name for future uses
13 df.rename(columns={"Insured Damage ('000 US$)": "Insured Loss"},
    inplace=True)
14
15 #Filtering data to accommodate values for the peril: Storm
16 storm_data = df[df['Disaster Type'] == 'Storm'].dropna(subset=['
    Insured Loss'])
17 severity = storm_data['Insured Loss'].values
18
19 #Setting the threshold to the 95th percentile (common)
20 u = np.percentile(severity, 95)
21 print(f"Threshold as 95th percentile: {u:.0f}")
22
23 #Finding exceedances above the threshold
24 exceedances = severity[severity > u] - u
25
26 #GPD fit
27 shape, loc, scale = genpareto.fit(exceedances)
28 print(f"GPD Fit: shape={shape:.4f}, scale={scale:.2f}")
29
30 #Cell 3
31 thresholds = np.linspace(min(severity), np.percentile(severity, 95)
    , 50)
32 mean_residual = [np.mean(severity[severity > th] - th) for th in
    thresholds]
33
34 plt.plot(thresholds, mean_residual, marker='o')
35 plt.xlabel('Threshold (US Dollars)')
36 plt.ylabel('Mean residual/excess')
37 plt.title('Mean Residual Life plot')
38 plt.grid(True)
39 plt.show()
40
41 #Cell 4
42 x = np.linspace(0, exceedances.max(), 100)
43 fitted_pdf = genpareto.pdf(x, shape, loc, scale)
44
45 plt.hist(exceedances, bins=30, density=True, alpha=0.6, label='
    Empirical Exceedances')
46 plt.plot(x, fitted_pdf, 'r', label='Fitted GPD')
47 plt.xlabel('Excess Loss')
48 plt.ylabel('Density')
```

```

49 plt.title('GPD Fit to Exceedances')
50 plt.legend()
51 plt.show()
52
53 #Cell 5
54 qqplot(exceedances, dist=genpareto, distargs=(shape,), loc=loc,
55         scale=scale, line='45', fit=True)
56 plt.title('Q-Q plot: GPD Fit')
57 plt.grid(True)
58 plt.show()
59
60 #Cell 6
61 if shape < 1:
62     expected_sev = scale / (1 - shape)
63     print(f"Expected Severity beyond {u}: {expected_sev:.2f}")
64 else:
65     print("Mean excess severity is infinite (Heavy-tailed
66           distribution)")

```

[language=Python]